

DSA Top-K Indexer: Process Write-Up (4-pager)

Author: Yury Kirpichev / Team Wombat Track: `dsa_topk_indexer_fp8_h64_d128_topk2048_ps64` Submission tags: `submission-v1` through `submission-v11` Final solution: `dsa_topk_indexer_fp8_b200_v3` (commit `31b8f71`)

Headline Summary

Field	Value
Hardware	NVIDIA B200 / <code>sm_100a</code> (FP8 E4M3 x FP8 E4M3 -> FP32, FP16 logits)
Workloads	128 / 128 PASSED
vs naive ref	964x mean (<i>unconfirmed — see caveat</i>)
vs FlashInfer/DG	38.4x mean (8.3x worst, 72.3x best) (<i>unconfirmed — see caveat</i>)
Aggregate latency	mean 7.83 us, p50 2.40 us, p95 17.80 us (<i>unconfirmed — see caveat</i>)
Source files	<code>solution/python/solution.py</code> , <code>solution/python/kernel.cu</code> , <code>solution/python/tcgen05_ptx.h</code> , <code>solution/python/umma_desc.h</code>

Caveat — comparison numbers are unconfirmed. The headline speedups (38.4x vs FlashInfer, 964x vs naive) come from `reports/submission-v10.md` / `reports/submission-v10-raw-cupti.log`. It is unclear at the time of writing whether the comparison numbers were end-to-end `cupti-python`-timed under the same harness configuration the official evaluator uses; some relative numbers may mix `cupti` and CUDA-event timings and will be remeasured.

Problem and Constraints

For each batch row `b`, the operator computes FP8 dot products `S[h] = sum_d Q[b,h,d] * K[page,slot,d]` over 64 heads / 128 dims with FP32 accumulation, weighted ReLU logits `logit[b,t] = scale[page,slot] * sum_h ReLU(S[h]) * weights[b,h]` for every position in `[0, seq_lens[b])`, top-K = 2048 selection, and a `block_table` transform back to physical page slots. The KV cache is FP8 with per-slot FP32 scales (132 bytes/slot, 64 slots/page), and outputs are pre-allocated (DPS).

The 128 contest workloads span batch 1-16 and `max_num_pages` 5 to 95+, meaning the smallest workloads have ~320 tokens (host launch overhead and TMEM allocation dominate) while the largest exceed 6000 tokens (HBM K-bandwidth and the TMEM-to-register readout dominate). A single kernel cannot serve both regimes well — dispatch by shape is mandatory.

Development was experiment-driven: each submission was benchmarked on Modal B200 with `cupti-python`, and changes were kept or reverted on measured numbers.

Challenges: Evaluation Framework (Pre vs Post PR #354)

The single largest constraint on the timeline was the evaluator. Before [flashinfer-bench PR #354](#) (merged Apr 10, 2026) there was no dedicated `DsaTopkIndexerEvaluator`. The default evaluator compared **raw output index vectors** element-wise, with two consequences: (1) when multiple positions tie on logit value any custom top-K — radix select, CUB sort, atomic emit — produced a valid top-K set that the evaluator rejected as `INCORRECT_NUMERICAL` because the index ordering did not match the reference's tie-breaking; and (2) the evaluator's random `k_index_cache_fp8` inputs were not packed in the deepgemm FP8 format the FlashInfer baseline expected, causing false negatives against FlashInfer itself. In practice the only top-K path that reliably passed was `torch::topk`.

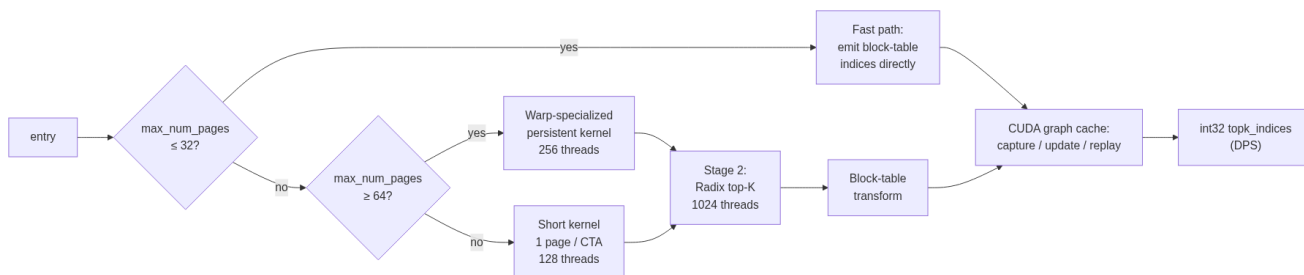
Pre-#354 (submissions v1-v4, Mar 29 - Apr 6). The early kernel was constrained to ATen-level ops that preserved bit-exact index agreement: a fused FP8 page-gather + dequant CUDA kernel, then `torch::bmm` for $Q \cdot K$, ATen `relu` + weighted sum, and `torch::topk`. Several more aggressive ideas were tried and reverted because they broke index-level agreement even when the selected values were correct: fused $Q \cdot K$ dot products, per-row scoring (no `[B,H,S]` materialization), BF16 reduction paths, chunked GEMM with running top-K. The ceiling under this constraint was $\sim 7.4x$ vs the naive reference.

Post-#354 (Apr 11 onwards). PR #354 introduced `DsaTopkIndexerEvaluator` which compares **sorted value vectors**, vectorizes index validation (duplicates, out-of-range, block-table reachability), and provides a `build_baseline` with correct FP8 packing. This unblocked the entire custom-kernel path. Within ten days the solution was rewritten from scratch: pure PyTorch FP8 dequant + bmm indexer (Apr 11-12), then custom CUDA top-K with CUB radix sort and FP8 MMA logits via `mma.sync.m16n8k32.e4m3` PTX (Apr 19-20), then a full SM100a UMMA rewrite with `tcgen05.mma.cta_group::1`, radix-select top-K, and size-aware dispatch (`v7`, `v8`, ~ 16 us mean), then persistent CTAs, warp specialization, Stage-2 SMEM cache, dispatch retuning and a CUDA graph cache (`v9` - `v11`). The final 7.83 us mean / 38.4x vs FlashInfer was reached on Apr 24.

In summary, the pre-#354 evaluator capped the work at $\sim 7x$ for six weeks; the post-#354 evaluator enabled the full custom CUDA pipeline that delivered 38.4x in the remaining two weeks.

Final Kernel Design

The kernel is a host dispatcher over three Stage-1 variants plus a fast-path bypass, a Stage-2 radix top-K, and a hand-rolled CUDA graph cache wrapping the whole launch sequence.



Fast path (`max_num_pages <= 32`). When the entire paged context fits within $K=2048$ every position is in the top-K, so the output is just block-table-transformed `[0, seq_len)` padded with -1, independent of Q , K , weights. 128 threads with vectorized int4 stores, ~ 2.3 us including graph replay overhead.

Short kernel (`33 <= max_num_pages < 64`). One CTA per page, 128 threads. Q and K loaded into 128B-swizzled SMEM, TMEM allocated, UMMA issued with 4 K -iterations on the $(M=128, N=64, K=32)$ shape,

FP32 accumulator read back, ReLU-weighted logits computed, FP16 emitted. Simple grid layout, minimal setup.

Warp-specialized persistent kernel (`max_num_pages >= 64`). 256 threads = producer warpgroup (warps 4-7, 40 regs, drives a 3-stage `cp.async K` pipeline) and math warpgroup (warps 0-3, 232 regs, issues UMMA, reads TMEM, computes ReLU/weighted sum, emits). `Q` is loaded once and TMEM allocated once per CTA. Per-stage `K_ready[buf] / K_done[buf]` mbarriers drive the handoff. A subtle bring-up bug: `cp.async.wait_group` is per-thread, so a producer-warpgroup `bar.sync` is required before the single-thread `mbarrier.arrive` on `K_ready[buf]` to ensure all threads' `cp.async` writes are visible. Grid is `(num_splits, B)` with `tiles_per_cta` tile-pairs per CTA.

Stage-2 radix top-K. Two-round radix select over 16-bit ordered keys finds the pivot in $O(\text{seq_len})$. Since `seq_len <= 16384` for all benchmarked workloads, ordered keys are cached in a 32 KB SMEM array on round 0 and re-read on subsequent passes (4x fewer HBM reads). Elements greater than the pivot are emitted first, equal-to-pivot fills remaining slots, then the block-table transform runs.

CUDA graph cache. `do_bench` clones inputs every iteration so pointers change but shapes (and the dispatch plan) stay fixed. For small workloads where the kernel itself is 2-10 us, ~2.5 us per `cudaLaunchKernel` dominated. The cache is a 32-slot LRU keyed by `(stream, dispatch_path, B, max_num_pages, tiles_per_cta, num_splits, max_len)`. Each call captures a fresh graph with current pointers and tries `cudaGraphExecUpdate` against the cached exec; on cache miss `cudaGraphInstantiate` creates a new exec. Falls back to direct launch if `cudaStreamBeginCapture` fails. This collapsed the fast-path bucket from 6.5 us to 2.3 us mean.

Negative Results and the Critical Path

The most useful finding was that the per-tile critical path in the persistent kernel is dominated by the TMEM-to-register transfer (`tcgen05_ld_32x32b_x64_b32 + tcgen05_wait_ld`), **not** by MMA compute, producer-prefetch latency, or the FMA accumulation chain. The following levers were measured and rejected: TMEM double-buffer + MMA-issue overlap (+3.6-5.2%, MMA not on the critical path); kKVStages 3 -> 4 (flat, producer not the bottleneck); 4-way accumulator split (within noise, FMA chain already hidden behind `wait_ld`); block-scaled UMMA `mx8f6f4` (infeasible, the data format uses per-row FP32 scales not per-K-block UE8M0); host-side `seq_lens.max()` sync (+20 us — `cudaStreamSynchronize` alone costs ~15-20 us on B200). Every micro-optimization that added work to the post-`wait_ld` emit path (in-Stage-1 ordered-u16 conversion, fused block-table transform in emit, skipped pad writes) regressed the 40-63 page bucket by 0.6-3.7% by lengthening the critical path between successive TMEM loads. The only structural way to reduce `wait_ld` cost is to shrink `kUMMA_N` via 2-CTA cluster UMMA, which was deferred due to infrastructure complexity (new PTX wrappers, cluster launch, DSMEM, cluster-scoped mbarriers). Phase 3 micro-optimizations that did stick: a Stage-2 ordered-u16 SMEM cache (4x fewer HBM reads), int4-vectorized fast-path stores, and re-tuning the persistent-kernel dispatch threshold from `max_num_pages >= 40` to `>= 64` (recovers 8.9% on the 40-63 page bucket where the short kernel's grid-per-tile layout saturates B200's 148 SMs faster than the persistent pipeline).

Validation

All 128 workloads pass the post-#354 `DsaTopkIndexerEvaluator`. Final benchmarks were collected on Modal B200 with `cupiti-python` matching the official evaluator's harness (warmup 3, iterations 100, 5 trials).

Tools and Languages

Tool / Language	Role
CUDA C++ (<code>sm_100a</code>)	All kernels: UMMA via inline PTX, <code>cp.async</code> pipelines, radix top-K
<code>torch.utils.cpp_extension.load()</code>	JIT compilation with <code>-gencode arch=compute_100a,code=sm_100a</code>
Inline PTX (<code>tcgen05_ptx.h</code>)	TMEM alloc/dealloc, UMMA issue/commit/wait, <code>cp.async</code> , <code>mbarrier</code> , <code>warpgroup</code> reg reconfig
CUTLASS bitfield layouts (<code>umma_desc.h</code>)	<code>SmemDescriptor</code> and <code>InstrDescriptor</code> for UMMA, vendored for self-containment
Python (<code>solution.py</code>)	Thin wrapper: JIT-compile and calls the extension
Modal	Cloud B200 access for benchmarking
<code>cupti-python</code>	GPU-side timing (matches official evaluation methodology)
Cursor + LLM agents	Agent-assisted development (see disclosure below)

References

Submitted source

1. Final kernel implementation - [solution/python/kernel.cu](#).
2. Python entry point and JIT compile flags - [solution/python/solution.py](#).
3. Inline PTX wrappers for SM100a - [solution/python/tcgen05_ptx.h](#).
4. UMMA descriptor layouts (vendored from CUTLASS) - [solution/python/umma_desc.h](#).
5. Submission manifest - [config.toml](#).

Contest documentation

1. Contest rules - [README.md](#).
2. FAQ (allowed optimizations, `torch.utils.cpp_extension`, custom compile flags) - [FAQ.md](#).
3. Evaluation environment - [EVALUATION.md](#).

Performance artifacts

1. Final cupti-timed comparison report - [reports/submission-v10.md](#).
2. Raw cupti-timed log - [reports/submission-v10-raw-cupti.log](#).
3. Optimization history and ablation results - [IMPROVEMENTS.md](#).
4. Future ideas and deferred optimizations - [experiments/FUTURE_IDEAS.md](#).

External references

1. Flashinfer-bench PR #354: `DsaTopkIndexerEvaluator` for correct tie-breaking comparison - [github.com/flashinfer-ai/flashinfer-bench/pull/354](#).
2. FlashAttention-4 (Dao et al., arXiv 2603.05451, Mar 2026) - asymmetric scaling, 2-CTA MMA, TMEM intermediates, warp specialization.
3. NVIDIA CUTLASS Blackwell docs - `tcgen05.mma.kind::f8f6f4`, `cta_group=2`, `SmemDescriptor`/`InstrDescriptor` layouts.

4. Colfax Research - "Writing GEMM Kernels Using Tensor Memory For Blackwell GPUs."
5. DeepGEMM `sm100_fp8_paged_mqa_logits.cuh` - reference production kernel with warp specialization, TMA pipelines, persistent scheduling.
6. NVIDIA CUDA C++ Programming Guide, CUDA graphs and `cudaGraphExecUpdate` - docs.nvidia.com/cuda/cuda-c-programming-guide/index.html#cuda-graphs.
7. PyTorch JIT C++/CUDA extensions, `torch.utils.cpp_extension.load` - pytorch.org/docs/stable/cpp_extension.html.
8. FlashInfer attention library (baseline used by the contest harness) - github.com/flashinfer-ai/flashinfer.

AI / Agent-Assisted Development Disclosure

Agent-assisted (Cursor + Claude/GPT-family LLMs); direction and acceptance criteria were human, all changes were accepted on measured B200 numbers. The agent was used as a pair-programming and refactoring tool for code generation, inline PTX wrapper authoring, and experiment scaffolding. All design decisions (kernel architecture, dispatch thresholds, pipeline depth, optimization accept/reject) were made by the human author based on benchmark measurements. The agent never had access to contest workload data or evaluation results beyond what was visible in the development logs.